

Flat parts and assembly limits

C.6 / P.3 PROTOTYPE REVIEW - PHYSICAL / BENCH QUALIFICATION PENDING

R1-R43

Existing Yageo MFR-25 values

All horizontal, 10.16 mm lead pitch

C1,C3,C5,C13,C17,C25

Panasonic ECEA1CKA101

100 uF / 16 V; max flat height 7.3 mm*

C9,C11,C15,C21

Panasonic ECEA1CKA100

10 uF / 16 V; max flat height 5.0 mm*

Q1-Q6

Existing exact transistor MPNs

Flat marked face up; preserve pin order

* Manufacturer maximum can diameter plus 0.5 mm insulating support; not scaled artwork.

Both capacitor types use the full 8 mm maximum body-length courtyard.

Capacitor square pad 1 is POSITIVE. Sleeve stripe marks NEGATIVE pad 2.

10 uF native lead pitch is 1.5 mm: form to the 2.54 mm PCB pitch without crossing.

Support leads at the seal while bending; do not pull, twist, or stress the rubber seal.

Use nonconductive support under horizontal cans. Do not obstruct the pressure vent.

TO-92 bodies extend toward the footprint body rectangle; marked flat face stays up.

Q6 is LP0701N3-G, not BC327. Verify actual pinout before forming or soldering.

IR lenses must still face the case opening; optical parts are not flattened blindly.

Sockets, screen, buttons and the front RGB retain their actual functional height.

The socketed ESP32 rear envelope is about 15.3 mm; it, not the caps, limits spacing.

The socketed RGB front envelope remains about 22.1 mm. Case depth is NOT final.

Power board remains a factory-assembled SMT subassembly, not a student SMT kit.

Prototype checks still required

Measure delivered screen, ESP32, SD module, sockets, plugs and standoffs.

Check horizontal capacitor ripple/rail transients during Wi-Fi, audio and motor bursts.

Check component temperatures, charging, rail sequencing and battery protection.

Confirm antenna performance with the actual rear board, battery, harness and case.

Prototype files issued separately; approve physical fit and factory DFM before ordering.